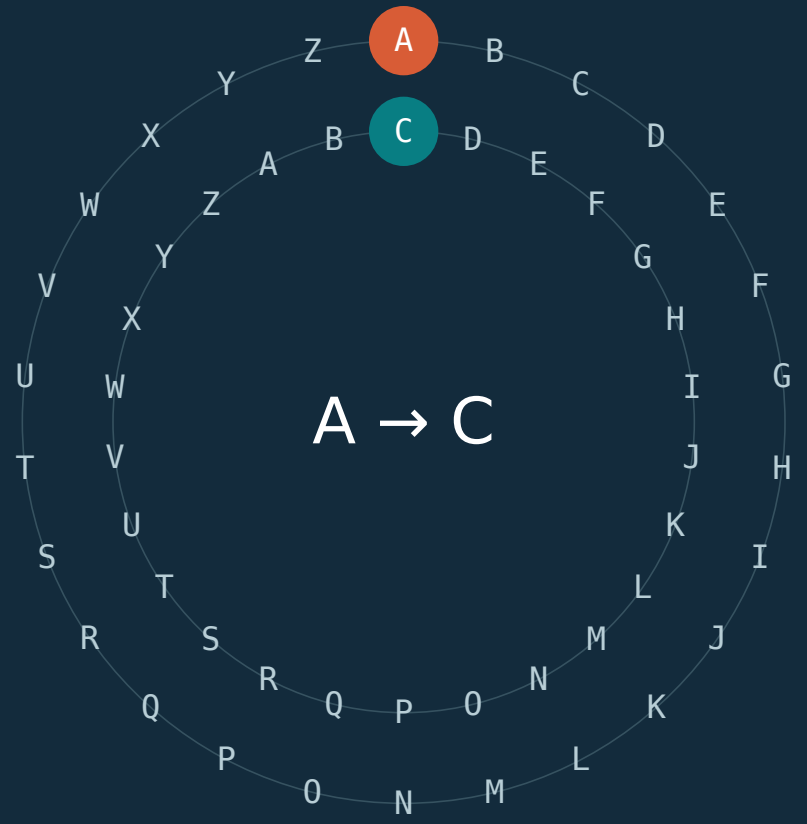
Cryptography lecture series

Classical ciphers

Historical ciphers and cryptanalysis



FOUNDATIONS

Today: constructing and analysing ciphers

Lecture 00 surveyed confidentiality, integrity, and authentication. Here we focus on **confidentiality**: keeping the message private.

Cryptanalysis studies how a proposed protection can be defeated under specified assumptions.

For each cipher, we will define the construction, work an example, and investigate what information it reveals.

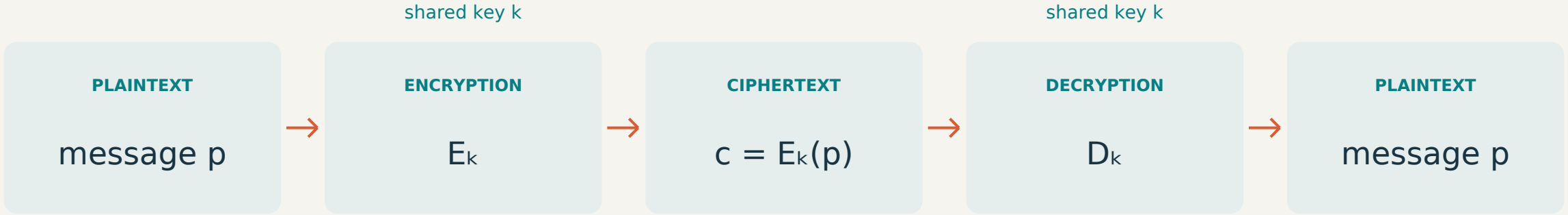
From everyday applications to mathematical constructions and their analysis.

Plaintext, ciphertext, algorithm, key

Term	Meaning in today’s setting
Plaintext	The message before encryption.
Ciphertext	The transformed message sent over the channel.
Cipher / algorithm	The rule used to encrypt and decrypt.
Key	The parameter that selects a particular transformation.

Knowing a ciphertext is different from knowing the key that produced it.

Encryption and decryption



Here Alice and Bob already share k. The diagram names plaintext p, ciphertext c, encryption E_k, and decryption D_k.

Symmetric-key encryption

Lecture 00’s browser example established a protected connection without a pre-shared secret.

Today’s starting assumption: Alice and Bob already share a secret key.

All five classical ciphers here are **symmetric-key** constructions: that shared key determines both encryption and decryption.

Key distribution is a separate problem; we assume it is solved for these examples.

A mathematical model of a cipher

Specify a plaintext space, a ciphertext space, and a key space.

$$\mathcal{P}, \quad \mathcal{C}, \quad \mathcal{K}; \qquad E_k : \mathcal{P} \to \mathcal{C}, \quad D_k : \mathcal{C} \to \mathcal{P}$$

Correctness means that decryption recovers every allowed plaintext:

$$D_k(E_k(p)) = p \qquad \text{for every } p \in \mathcal{P} \text{ and } k \in \mathcal{K}.$$

We will identify these objects explicitly for each construction.

What does the attacker know?

Assume Eve knows the cipher algorithm, but not the particular key.

Information available	Attack setting
Only intercepted ciphertext	Ciphertext-only
Some matching plaintext and ciphertext	Known-plaintext
Encryptions of messages selected by Eve	Chosen-plaintext

Our main substitution and Vigenère attacks use ciphertext-only information.

Public-algorithm principle: Kerckhoffs, La Cryptographie Militaire (1883).

The alphabet as a mathematical object

LETTER ENCODING

A	B	C	D	E	…	X	Y	Z
0	1	2	3	4	…	23	24	25

$$\mathbb{Z}_{26} = \{0, 1, \ldots, 25\}, \qquad \mathcal{P} = \mathcal{C} = \mathbb{Z}_{26}^{n}$$

Normalize examples to letters only. We use uppercase for message rows; partial substitution solutions use lowercase.

Removing spaces or merging letters is preprocessing, and may discard information.

Modulo 26: the alphabet wraps around

$$(24 + 2) \bmod 26 = 0, \qquad (0 - 2) \bmod 26 = 24$$

LETTER INTERPRETATION

`Y + 2 → A` `A − 2 → Y`

We represent every residue by an integer from 0 to 25.

When using indexed messages, positions start at 0. Permutation tables later explicitly use positions 1 through m.

Historical motivation and our route

Construction	Historical / mathematical idea
Shift	A small offset makes hand encryption simple.
Substitution	An arbitrary alphabet gives many more keys.
Vigenère	Change the alphabet with the key position.
Playfair	Encrypt pairs using a compact square.
Transposition	Rearrange positions instead of changing letters.

We will understand each construction before examining its limitations.

SHIFT

01 / Shift cipher

Use one fixed offset throughout the message.

A familiar historical example is Caesar’s shift by three positions.

The offset is the key; the alphabet and rule are public.

Shift cipher: definition

$$\mathcal{P} = \mathcal{C} = \mathbb{Z}_{26}^{n}, \qquad \mathcal{K} = \mathbb{Z}_{26}$$

$$c_i = E_k(p)_i = (p_i + k) \bmod 26$$

$$p_i = D_k(c)_i = (c_i - k) \bmod 26$$

There are 26 mathematical keys. The key 0 is the identity, leaving 25 nontrivial shifts.

Why does decryption work?

Encryption adds the same residue that decryption subtracts.

$$D_k(E_k(p))_i = ((p_i + k) - k) \bmod 26 = p_i$$

WRAP-AROUND CHECK

```
Y → A with k = 2
A → Y when we subtract 2
```

The equality is in the residue class ring; our displayed values are its representatives.

Shift example: compute the first letters

PLAINTEXT AND KEY

CRYPTOGRAPHYISFUN k = 2

Plaintext	Numeric calculation	Ciphertext
C = 2	(2 + 2) mod 26 = 4	E
R = 17	(17 + 2) mod 26 = 19	T
Y = 24	(24 + 2) mod 26 = 0	A
P = 15	(15 + 2) mod 26 = 17	R

Compute before revealing the complete row on the next slide.

Shift example: finish and reverse

ALIGNED MESSAGE ROWS

```
Plaintext   CRYPTOGRAPHYISFUN
Ciphertext  ETARVQITCRJAKUHWP
```

$$(4-2) \bmod 26 = 2\ (C), \qquad (0-2) \bmod 26 = 24\ (Y)$$

Apply the same subtraction to every ciphertext symbol to recover the normalized plaintext.

Recovering a shift key

Candidate k	Decryption of KHOORZRUOG
1	JGNNQYQTNF
2	IFMMPXPSME
3	HELLOWORLD

At k = 3, HELLOWORLD is meaningful, so we stop. There are at most 26 candidate keys to try.

$$\text{Known } p_i, c_i \quad \Longrightarrow \quad k = (c_i - p_i) \bmod 26.$$

Exercise: recover the message

Individual calculation • 2 minutes

INTERCEPTED

```
PHHWPHDIWHUFODVV
```

Find a plausible key and plaintext. Explain the evidence for your choice.

If you are told M encrypts to P, recover the key directly.

An answer and calculation are in the instructor guide.

02 / Substitution cipher

Replace the fixed shift by an arbitrary permutation of the alphabet.

Every occurrence of one plaintext letter still receives the same ciphertext letter.

Historical appeal: a much larger family of secret alphabets.

Substitution: a bijection and its inverse

$$\mathcal{P} = \mathcal{C} = \mathbb{Z}_{26}^{n}, \qquad \mathcal{K} = S_{26}$$

S_{26} is the set of all alphabet permutations. The secret key σ (sigma) is one chosen permutation: a one-to-one reassignment of the 26 symbols.

$$c_i = \sigma(p_i), \qquad p_i = \sigma^{-1}(c_i)$$

Its inverse σ^{-1} maps each ciphertext symbol back to its original plaintext symbol.

$$\sigma^{-1}(\sigma(p_i)) = p_i.$$

A one-to-one map is essential: a collision would prevent unique decryption.

Substitution example: use the table

INPUT ALPHABET

ABCDEFGHIJKLMNOPQRSTUVWXYZ

OUTPUT ALPHABET

QWERTYUIOPASDFGHJKLZXCVBNM

Input letters	Look up each image	Result
B E C A U S E	W T E Q X L T	WTEQXLT

Decrypt W by finding it in the output row: it lies under B.

Counting keys and preserving patterns

$$|\mathcal{K}| = 26 \cdot 25 \cdots 2 \cdot 1 = 26! \approx 4.03 \times 10^{26} \approx 2^{88.4}$$

Trying the entire space at one billion keys per second takes approximately 12.8 billion years.

$$p_i = p_j \quad \Longleftrightarrow \quad \sigma(p_i) = \sigma(p_j)$$

A PATTERN THAT SURVIVES

LETTER → STZZTK

A large key space rules out one simple attack; it does not rule out every attack.

Frequency analysis: why counting helps

A substitution changes the labels of the letters, but not their occurrence counts.

$$f_C(\sigma(a)) = f_P(a), \qquad \frac{f_C(\sigma(a))}{N} = \frac{f_P(a)}{N}$$

If the language has characteristic frequencies, the ciphertext inherits them under unknown labels.

Historical context: al-Kindi described frequency-based cryptanalysis in the 9th century.

A language model is a guide

Letter	E	T	A	O	I	N
Illustrative frequency	12.7%	9.1%	8.2%	7.5%	7.0%	6.7%

These are approximate English frequencies, not exact properties of every message.

Pairs such as TH, HE, IN, ER and triples such as THE, AND, ING provide additional evidence.

Absence of a common pair in a short message is weak evidence, not a logical impossibility.

Our intercepted message

168 LETTERS • ORIGINAL REFERENCE EXAMPLE

```
YIFQFMZRWQFYVECFMDZPCVMRZWNMDZVEJBTXCDDUMJ
NDIFEFMDZCDMQZKCEYFCJMYRNCWJCSZREXCHZUNMXZ
NZUCDRJXYYSMRTMEYIFZWDYVZVYFZUMRZCRWNZDZJJ
XZWGCHSMRNMDHNCMFQCHZJMXJZWIEJYUCFWDJNZDIR
```

Assume English plaintext. The unknown substitution key differs from the earlier illustrative table.

We will recover the message incrementally, keeping a partial key and checking each hypothesis.

1 / Count letters and propose a mapping

Ciphertext letter	Z	M	C	D	F	J
Count	20	16	15	13	11	11
Frequency	0.119	0.095	0.089	0.077	0.065	0.065

Z is the most frequent symbol: 20/168 ≈ 0.119. English e is often frequent, so begin by testing Z → e.

Other common letters remain possible. Each proposed mapping must fit every occurrence.

Counts suggest hypotheses; they do not identify letters with certainty.

2 / Combine pair clues to extend the key

Evidence	Reasoning under the current hypotheses
ZW: 4; WZ: 0	e? occurs repeatedly; ?e does not. W has frequency 0.0476. Test W → d.
RW: 2	?d suggests nd or ed. If Z → e is retained, bijectivity rules out R → e; try R → n.
NZ: 3; ZN: 1	HE is a common pair. Test N → h and inspect the resulting plaintext.

PROVISIONAL PARTIAL KEY

`Z → e  W → d  R → n  N → h`

The one-to-one constraint gives logical exclusions; language frequencies rank the remaining guesses.

3 / Read fragments and update the key

PARTIAL PLAINTEXT • · = UNDETERMINED

```
·····end·········e····nedh··e···········
h·····e····e·········nh·d···en····e·h··e
he···n·····n·····ed···e···e··ne·ndhe·e··
·ed····nh···h·····e····ed·····d··he··n
```

The fragment ne·ndhe suggests “…ne and he…”, adding C → a.

RNM becomes nh·, suggesting a vowel after h. CM becomes a·; ai is more plausible than ao. Test M → i.

Apply these candidates throughout the message; revise any that lead to contradictions.

4 / Use familiar pairs and word patterns

PARTIAL PLAINTEXT • · = UNDETERMINED

```
·····iend·····a·i·e·a·inedhi·e·····a···i·
h·····i·ea·i·e·a···a·i·nhad·a·en··a·e·hi·e
he·a·n·····in·i····ed···e···e·ineandhe·e··
·ed·a··inhi··hai··a·e·i··ed·····a·d··he··n
```

Clue	Candidate reading	Mapping to test
JN → ·h	th	J → t
XYYSMRT → ····in·	looking	Y → o
MD occurs four times: i·	is	D → s

The partly decoded HNCMF then reads ·hai·; chair proposes H → c and F → r.

5 / Summarize the remaining recovery

PARTIAL PLAINTEXT • · = UNDETERMINED

```
o·r·riend·ro··arise·a·inedhise··t···ass·it
hs·r·riseasi·e·a·orationhadta·en··ace·hi·e
he·asnt·oo·in·i·o·redso·e·ore·ineandhesett
·ed·ac·inhischair·aceti·ted··to·ardsthes·n
```

Recognizable fragments now suggest “our friend”, “from Paris”, and “looking”. These extend the key consistently.

The remaining words—examined, empty, with, evaporation, back—supply the last observed letters. Apply each addition across the whole message.

The repeated trial-and-check procedure is the same; we now move to the complete plaintext.

The recovered message

Our friend from Paris examined his empty glass with surprise, as if evaporation had taken place while he wasn’t looking. I poured some more wine and he settled back in his chair, face tilted up towards the sun.

OBSERVED INPUT LETTERS

ABCDEFGHIKLMNOPRSTUVWXY

THEIR CIPHERTEXT IMAGES

CGHWZQTNMSXVRYEFDJIKUPB

J, Q, Z are absent: 3! = 6 full substitution tables still extend the 23 observed mappings.

Original reference example; spacing and punctuation restored by interpretation.

What the attack has taught us

We combined counts, pair statistics, repeated fragments, and plausible word boundaries.

Some decisions followed from the bijection constraint; others were statistical or contextual hypotheses.

A successful reconstruction explains all observed positions with one consistent substitution.

Exercise: identify one logical exclusion and one probabilistic guess in the walkthrough.

BREAK 01

Break / 10 minutes

We have seen how a fixed substitution preserves the language’s structure.

Next we change the substitution with position and ask which patterns remain.

7:40–7:50 pm • Resume at 7:50 pm

03 / Vigenère cipher

Use several shift substitutions, selected by a repeating keyword.

Repeating-key polyalphabetic methods were described by Bellaso in 1553; the familiar name is associated with Vigenère.

Historical objective: weaken the direct link between letter identity and frequency.

Vigenère: spaces and key schedule

$$\mathcal{P} = \mathcal{C} = \mathbb{Z}_{26}^{n}, \qquad \mathcal{K} = \mathbb{Z}_{26}^{m}$$

$$k = (k_0, \ldots, k_{m-1}), \qquad c_i = (p_i + k_{i \bmod m}) \bmod 26$$

$$p_i = (c_i - k_{i \bmod m}) \bmod 26$$

The keyword has m letters. Its effective repeating period can be smaller, as in ABCABC.

Encryption example: repeat the keyword

WRITE THE ALIGNMENT FIRST

```
Plaintext   CRYPTOGRAPHY
Keyword     LUCK
Key stream  LUCKLUCKLUCK
```

Position i	0	1	2	3	4	5
Plaintext value	2	17	24	15	19	14
Key value	11	20	2	10	11	20

Predict the first six ciphertext letters before continuing.

Encryption example: do the arithmetic

i	p_i	Key value	Sum mod 26	Ciphertext
0	2	11	13	N
1	17	20	11	L
2	24	2	0	A
3	15	10	25	Z
4	19	11	4	E
5	14	20	8	I

The wrap-around at i = 1 gives (17 + 20) mod 26 = 11, or L.

Complete the example and decrypt

COMPLETE ROWS

```
Plaintext   CRYPTOGRAPHY
Key stream  LUCKLUCKLUCK
Ciphertext  NLAZEIIBLJJI
```

$$p_1 = (11 - 20) \bmod 26 = 17 \quad (R)$$

Decryption subtracts the same aligned key value. Adding and subtracting cancel at each position.

R at positions 1 and 7 encrypts to L and B, because the aligned key letters differ.

Why does ordinary frequency analysis weaken?

A plaintext letter can be shifted in several different ways, depending on its position in the key schedule.

The aggregate histogram mixes these shifted distributions.

However, positions sharing the same key index still use one fixed shift.

The attack will isolate those positions instead of treating all letters as one population.

Cryptanalysis has two linked stages

Stage	Goal	Evidence / method
1	Find plausible periods	Repeated segments; a letter-match check called the index of coincidence
2	Recover the shifts	Frequency counts within aligned columns

Validate the combined result by decrypting the message. A plausible period alone does not identify the key.

Kasiski: see the repeated alignment



The highlighted THE groups begin 8 positions apart. The four-letter key KING has completed two cycles, so both groups encrypt to BUK.

Repeated ciphertext segments can suggest a key period; accidental repetitions are also possible.

Kasiski: turn repetition into a candidate length

Now consider a longer intercepted message. FVR repeats at the positions below.

Repeated group	Position	Distance from first
FVR	97	—
FVR	147	50
FVR	552	455
FVR	897	800

$$\gcd(50, 455, 800) = 5$$

Try key lengths 1 and 5; compare their letter-match rates next.

Zero-based positions. Dataset: 1,045 letters; full ciphertext in Appendix A1–A2.

Index of coincidence (IC): a match check

The index of coincidence, or IC, is the probability that two randomly chosen distinct positions contain the same letter.

position 0	position 1	position 2
A	A	B

Pair of positions (0, 1)	Pair of positions (0, 2)	Pair of positions (1, 2)
A–A: match	**A–B: different**	**A–B: different**

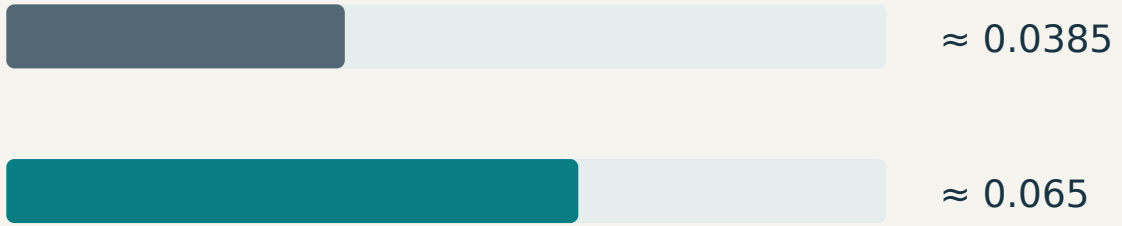
1 matching pair out of 3 → IC = 1/3

This tiny example explains the counting idea; its value is not a typical English reference value.

What does the match rate tell us?

In English, some letters occur much more often than others. This raises the chance that two sampled letters match.

REFERENCE MATCH RATES FOR LONGER TEXTS



A fixed shift renames the letters without changing their counts, so it preserves the match rate.

These are reference values for longer texts. Short samples and different languages can vary considerably.

A compact formula after the intuition

For a text of N letters, let f_a be the number of occurrences of letter a.

$$\mathrm{IC} = \frac{\Sigma_{a=0}^{25} f_a(f_a - 1)}{N(N-1)}$$

The numerator counts equal-letter pairs; the denominator counts all ordered pairs of distinct positions.

A program can perform this count for us. The derivation and a larger worked calculation are in the appendix.

Use IC as evidence about letter distributions, rather than a general security test.

Try five: group the same key positions

	Key slot 1	Key slot 2	Key slot 3	Key slot 4	Key slot 5
Positions 0–4	E	L	Q	V	V
Positions 5–9	D	X	A	F	L
Positions 10–14	Z	J	E	S	P
Positions 15–19	C	I	F	K	E
Positions 20–24	T	X	U	B	T

If the key repeats every five letters, reading down one colour collects letters encrypted with one fixed shift.

The colour labels mark key positions, not recovered key letters. The five columns each contain 209 letters.

Check the candidate length with IC

Candidate length	Observed match rate
1: keep all letters together	0.0429
5: five separate columns	See the individual values below.

Column	1	2	3	4	5
IC	0.0646	0.0679	0.0638	0.0679	0.0699

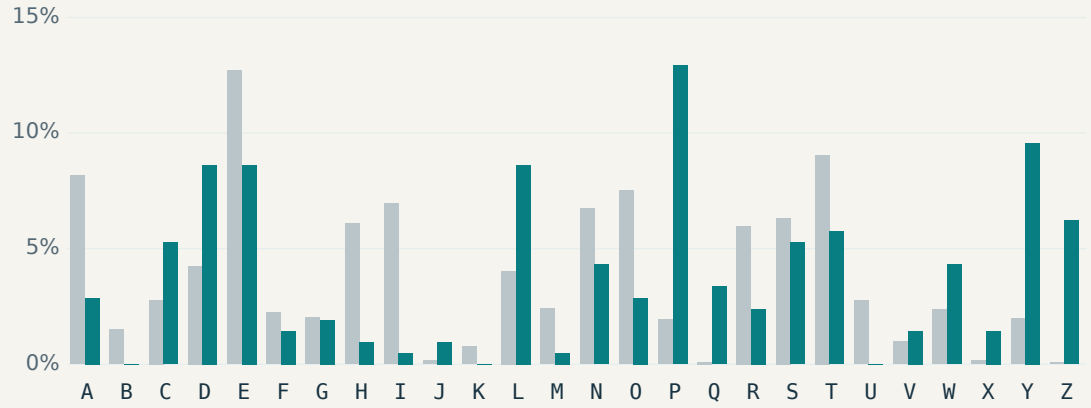
All five columns have English-like match rates. Together with the repeats, this supports key length 5.

A multiple such as 10 can also look plausible; a high IC does not uniquely determine the shortest period.

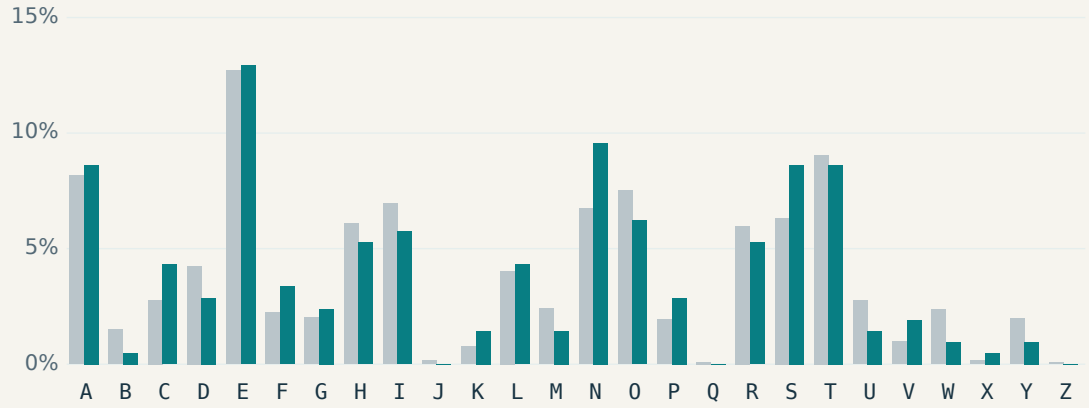
Recover one shift by comparing frequencies

In column 1, P occurs 27 times. If it represents E, the shift is 15 − 4 = 11, giving key letter L.

Subtract 0 (no shift)



Subtract 11 (candidate L)



English model

Trial-decrypted column 1 • compare all 26 letters

Try all 26 shifts and compare the whole frequency pattern. The detailed scoring method is in the appendix.

Finish the key and check the message

Column	Recovered shift	Key letter
1	11	L
2	4	E
3	12	M
4	14	O
5	13	N

REJOIN AND DECRYPT WITH LEMON

THEHISTORYOFSECRETWRITINGISALSOAHISTORY…

The coherent full plaintext supports the combined key hypothesis. Individual columns need not read as words.

The attack as a sequence of simple checks

Question	What we did
Where might the key repeat?	Measure distances between repeated ciphertext groups.
Does five look plausible?	Group aligned positions and check their IC values.
Which shifts fit the columns?	Compare each trial-decrypted distribution with English.
Does the whole message make sense?	Rejoin the columns and decrypt with LEMON.

Repeats → candidate length → match-rate check → shifts → readable message.

BREAK 02

Break / 10 minutes

We used repetitions, letter matches, and frequency patterns to recover a repeating key.

Next we replace single-letter substitution with pairs, then rearrange positions.

8:45–8:55 pm • Resume at 8:55 pm

04 / Playfair cipher

A compact construction for encrypting letter pairs by hand.

Invented by Charles Wheatstone in 1854 and promoted by Lyon Playfair.

Historical attraction: pair substitution without carrying a huge substitution table.

Why substitute more than one letter?

Single-letter substitution leaves each letter’s frequency directly visible under another label.

Playfair works with **digrams**: pairs of distinct letters prepared from the message.

Codebooks could replace entire words or instructions, but large books were harder to distribute and protect.

This is polygraphic substitution; Vigenère instead changes a single-letter mapping with position.

Build the square: normalize the keyword

KEYWORD

PLAYFAIR EXAMPLE

KEEP FIRST OCCURRENCES ONLY

P L A Y F I R E X M

Merge J into I, remove spaces, and scan left to right, skipping any letter already entered.

These ten distinct letters occupy the first two rows of the 5 × 5 square.

Build the square: fill the remaining cells

First enter the keyword letters.

P	L	A	Y	F
I	R	E	X	M

Then append unused letters in order.

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

Unused letters: B C D G H K N O Q S T U V W Z. Both parties build the same square.

Prepare the message, one pair at a time

Remaining text	Action	Pair emitted
BALLOON	B and A differ; consume both.	BA
LLOON	Repeated L: consume first L; insert X.	LX
LOON	L and O differ; consume both.	LO
ON	O and N differ; consume both.	ON

Thus BALLOON becomes BA LX LO ON. Do not consume both repeated L letters when inserting the filler.

Preparation conventions and the gold example

If one letter remains, append X. For a repeated or final X, use Q as the alternate filler.

EXAMPLES

`X → XQ` `XX → XQ XQ`

HIDE THE GOLD IN THE TREE STUMP

`HI DE TH EG OL DI NT HE TR EX ES TU MP`

Only letters within the next proposed pair trigger the repeated-letter rule.

Preparation can lose information; we keep it separate from the reversible pair transformation.

Rule 1: letters in the same row

1. Locate P and L in row 1.
2. Move each one cell to the right.
3. Preserve their order.

RESULT

```
PL → LA
Wrap-around: FP → PL
```

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

Grid shading marks the input pair. Decryption moves left instead of right.

Rule 2: letters in the same column

1. Locate P and I in column 1.
2. Move each one cell down.
3. Wrap from the bottom to the top.

RESULT

PI → IB

Wrap-around: TP → PI

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

Decryption moves up; both row and column rules wrap around the square.

Rule 3: opposite rectangle corners

H is at (row 3, column 5).

I is at (row 2, column 1).

Keep each input row; exchange the columns.

RESULT

```
H → B at (3,1)
I → M at (2,5)
HI → BM
```

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

The first output stays in the first input’s row; pair order matters.

Encrypt the prepared message in stages

Input pair	Relation in the square	Output
HI	Rectangle	BM
DE	Same column	OD
TH	Rectangle	ZB
EG	Rectangle	XD

Work through these first four pairs before applying the same rules to the rest.

Complete the encryption and reverse it

PREPARED PLAINTEXT

HI DE TH EG OL DI NT HE TR EX ES TU MP

CIPHERTEXT

BM OD ZB XD NA BE KU DM UI XM MO UV IF

For decryption: move left in a row, up in a column, or apply the same rectangle rule.

Check BM → HI and OD → DE using the square.

Standard PLAYFAIR EXAMPLE illustration, also present in the supplied deck.

Exercise: decrypt with the same square

Pair exercise • 3 minutes

CIPHERTEXT

```
RO NC FL WI
```

Decrypt each pair and name the rule used.

Keep any filler until you have recovered all the prepared letters.

P	L	A	Y	F
I	R	E	X	M
B	C	D	G	H
K	N	O	Q	S
T	U	V	W	Z

The complete calculation follows on the next slide.

Solution: show every pair and rule

Ciphertext	Inverse rule	Prepared plaintext
RO	Rectangle	EN
NC	Same column: move up	CR
FL	Same row: move left	YP
WI	Rectangle	TX

RESULT

```
EN CR YP TX → ENCRYPTX
```

The intended reading is ENCRYPT, with a final X supplied during preparation.

Why exact spelling may not be recoverable

TWO DIFFERENT ORIGINALS

```
AXA → AX AX
AA  → AX AX
```

Both originals produce the same prepared text and therefore the same ciphertext.

Merging I and J also loses information. Do not automatically remove every X from a decryption.

The pair transformation is invertible on prepared pairs; preprocessing need not be invertible.

Historical usefulness and cryptanalysis

Playfair offered a practical balance between manual operation and more complex substitution; it saw military field use.

Pair and longer language patterns still survive. A computer can decrypt with candidate squares and score the results.

HILL-CLIMBING OUTLINE

```
square → decrypt → language score
modify square → keep improvement → repeat
```

Random restarts help with local optima. Success depends on the message, language model, and search effort.

Understand the historical design goal before judging it by present-day requirements.

05 / Transposition cipher

Keep letter identities unchanged and permute their positions within each block.

Substitution changes symbols; transposition changes where they occur.

Permutation cipher: formal definition

$$\mathcal{P} = \mathcal{C} = \mathbb{Z}_{26}^{m}, \qquad \mathcal{K} = S_m$$

$$E_\pi(p_1, \ldots, p_m) = (p_{\pi(1)}, \ldots, p_{\pi(m)})$$

$$D_\pi(c_1, \ldots, c_m) = (c_{\pi^{-1}(1)}, \ldots, c_{\pi^{-1}(m)})$$

Apply this operation block by block. Positions here are explicitly numbered 1 through m.

For incomplete blocks, pad with X and retain the original message length separately.

Read the permutation in the right direction

Output position i	1	2	3	4	5	6
Input position π(i)	3	1	4	6	2	5

The i-th ciphertext letter comes from input position π(i).

FIRST THREE OUTPUTS

$c_1 = p_3 \qquad c_2 = p_1 \qquad c_3 = p_4$

Some sources define the reverse direction. This table fixes the convention used throughout our example.

Encrypt one block, step by step

Output position	1	2	3	4	5	6
Read input position	3	1	4	6	2	5
Input letter from ATTACK	T	A	A	K	T	C

TWO-BLOCK MESSAGE

```
ATTACK ATDAWN
TAAKTC DAANTW
```

Every letter appears exactly as often after rearrangement as before it.

Construct the inverse permutation

Original input j	1	2	3	4	5	6
Output position containing j	2	5	1	3	6	4

READ THE INVERSE

$\pi^{-1} = [2, 5, 1, 3, 6, 4]$

TAAKTC → ATTACK

$$c_{\pi^{-1}(j)} = p_{\pi(\pi^{-1}(j))} = p_j.$$

The equality explains why the inverse table undoes encryption.

Exercise: encrypt, then explain an invariant

Individual calculation • 2 minutes

PLAINTEXT AND KEY

DEFEND

$\pi = [3, 1, 4, 6, 2, 5]$

Encrypt the block and recover it using the inverse permutation.

How many keys are possible at block size 6? Which letter statistics are unchanged?

Use the declared output-to-input convention, not a memorized alternative.

Solution and security discussion

SOLUTION

DEFEND → FDEDEN → DEFEND

$$|S_6| = 6! = 720, \qquad f_C(a) = f_P(a) \text{ for every letter } a.$$

For this block size, exhaustive search is small. Candidate rearrangements can also be ranked using language patterns.

With larger blocks there are more keys, but unchanged counts remain an important structural property.

SYNTHESIS

Compare the five constructions

Cipher	Mathematical key	Information exploited in an attack
Shift	$k \in \mathbb{Z}_{26}$	Small key space
Substitution	$\sigma \in S_{26}$	Letter and word-pattern statistics
Vigenère	$k \in \mathbb{Z}_{26}^{m}$	Periodicity and column statistics
Playfair	A keyed 5×5 square	Pair and longer language statistics
Transposition	$\pi \in S_m$	Preserved counts and positional patterns

Each construction illustrates a different relationship between a transformation and its invariants.

SYNTHESIS

From historical alphabets to digital messages

Our hand-cipher examples first removed spaces and punctuation. That preprocessing prevented exact recovery of the original formatting.

Digital encryption operates on encoded bytes or bits and can recover the complete original representation.

The next lectures develop stronger constructions and additional security goals. These five ciphers establish the language and methods for that study.

Historical usefulness and present-day suitability are different questions.

APPENDIX

A1 / Complete Vigenère ciphertext

POSITIONS 0–527 • ZERO-BASED

```
ELQVVDXAFLZJESPCIFKETXUBTTWMZFZETWFESDMBQTQCCWIR
WAOMZUCLXFSEYWMGRYHQFZLCDSCWEOSRGIDMYPXFSEHMFVNY
SFVRCWKAOZPMBQSSBSGSEFHUPQQGFLKQKVWPDSZLMZVVOHQB
LPXFVRSENWGDSROYLRSINRIEIEGMHSVYWGFCCMEWARAMMFNS
YABYPQHGPVEGGTPXCPNYDCSEIZFRAIMHROAAFQDPQOIPVQDR
LXQRFSEBSFLRPTNXMXWNCTTFNDIEDEZZURRNPGSFLGMFRQYX
FRLHQFQZIEBBEXDSNEEZMFTRSZRNPGSNDTDCBQMZGGPEPHUP
VQOQPVYOXPWMUHPWEQUPGWGVEWOCADICIRYGQGNYHOVNYKQG
GSISIRDWIVRYXTSRGMPSANIPWFLKDSRDGAACFXQFFXEWSGSM
EDEZGQGFQEEHRCFGHQZRAHEPQAJRELQBRPHRCELGXSNCQARR
WSRHUPTDCOWIYKUPRMGUZVFYRJMEFRAIMHROEOFBDWMZBYKY
```

Join both pages without spaces or line breaks. All Kasiski positions and IC values refer to this 1,045-letter string.

APPENDIX

A2 / Complete Vigenère ciphertext

POSITIONS 528–1044 • ZERO-BASED

```
SFDESSGSIEOZPTAGVEMABFTRFVRVIKFREYDBNREUBNYHMUNT
REDYTXFWARXTSZPWEOTPMZHBNSXIZYWOOACIHSNWEEWZAPQD
NEXQFATREWQPIMQUNSXIZYIZCHRLFSKEKUJRDXTSNYEXMFEE
NSGEIDQULROSBQHUGGTRSIVDLUBTLYESSFPBOGEIDBSCSYOP
ZMZQVOIZQRELQZRDWABVDXTOGLPMFTPRGAOPVATCZWEWOWIW
SLDMECAWCABRAEDHBQWQQHCMFMJPQGGGLPECNDOIVNEMZTBC
QMHVZRDSZLMZGITWUPYPEZRJSEFOAZFESEGIDQNYPQOEYJDC
ZTXSCBOIZUVYIQFVYKEHNCXEPLDXMHVYKFVRLWEIZAXUCADG
TSPVMZURIEYDYPWMBQWSAYVYKRCEHEKGGSEFOFJWFSZNEZTN
TPMGRNYDSQPWUUAXYEHFFVHWIPGMFRQYXDHMPUQFEYPMELXT
SEELMBQPTQBQZRWSRAMZUGSIYSGSSPIAVRAKA
```

Join both pages without spaces or line breaks. All Kasiski positions and IC values refer to this 1,045-letter string.

APPENDIX

A3 / All counts for the first Vigenère column

Letter	Count	Letter	Count	Letter	Count	Letter	Count
A	6	B	0	C	11	D	18
E	18	F	3	G	4	H	2
I	1	J	2	K	0	L	18
M	1	N	9	O	6	P	27
Q	7	R	5	S	11	T	12
U	0	V	3	W	9	X	3
Y	20	Z	13				

These 26 counts sum to 209. Use all of them when computing IC or scoring candidate shifts.

APPENDIX

A4 / Illustrative English frequency model

Letter	q_a	Letter	q_a	Letter	q_a	Letter	q_a
A	0.08167	B	0.01492	C	0.02782	D	0.04253
E	0.12702	F	0.02228	G	0.02015	H	0.06094
I	0.06966	J	0.00153	K	0.00772	L	0.04025
M	0.02406	N	0.06749	O	0.07507	P	0.01929
Q	0.00095	R	0.05987	S	0.06327	T	0.09056
U	0.02758	V	0.00978	W	0.02360	X	0.00150
Y	0.01974	Z	0.00074				

Rounded for display. Calculations use normalized unrounded values; corpus frequencies vary.

APPENDIX

A5 / Recovered Vigenère prose, part 1

The history of secret writing is also a history of people finding patterns. A sender may replace every letter with another symbol and hope that the message will remain hidden. Yet the habits of a language survive in surprising ways. Common letters still occur often, repeated words leave repeated shapes, and familiar phrases provide clues. A careful reader does not treat any single clue as proof. Instead the reader makes a guess, checks its consequences, and changes the guess when the evidence disagrees. Computers make this process faster but do not remove the need for a clear model of the problem.

Word boundaries and punctuation are editorial reconstructions, not encoded by the cipher.

APPENDIX

A6 / Recovered Vigenère prose, part 2

When a short key is repeated across a long message, the same positions in the key return again and again. Splitting the message into columns can reveal a simple pattern inside each column. Enough text gives the analyst a better chance of distinguishing a useful pattern from a coincidence. The lesson is that a large number of possible keys is only one part of security. We must also ask what information remains visible and what an observer can learn from it. Good engineering starts by stating the assumptions, checking examples, and looking for ways that a system can fail. A secure design must survive careful public study rather than depend on keeping the method unknown.

APPENDIX

A7 / Deriving the finite-sample IC formula

There are N(N − 1) ordered pairs of distinct positions.

For a letter a, exactly $f_a(f_a - 1)$ pairs have a in both positions. These events are disjoint across letters.

$$\mathrm{IC}(X) = \frac{\sum_{a=0}^{25} f_a(f_a - 1)}{N(N-1)}$$

$$= \sum_{a=0}^{25} \frac{f_a}{N} \frac{f_a - 1}{N - 1}.$$

APPENDIX

A8 / A small IC calculation, fully worked

STRING

AABAAB

Letter	Count f_a	$f_a(f_a - 1)$
A	4	12
B	2	2
All others	0	0

$$\mathrm{IC}(\mathrm{AABAAB}) = \frac{12 + 2}{6 \cdot 5} = \frac{7}{15} \approx 0.4667.$$

Count matching pairs; do not divide by N^2 when positions must be distinct.

APPENDIX

A9 / From sample counts to language probabilities

If independent letters have probabilities q_0, ..., q_{25}, then the probability of equality is:

$$\Pr[X_i = X_j] = \sum_{a=0}^{25} q_a^2 \qquad (i \neq j).$$

Writing $\hat{q}_a = f_a/N$ gives an exact relation for the observed sample:

$$\mathrm{IC}(X) = \frac{N \, \Sigma_a \, \widehat{q}_a^{\,2} - 1}{N - 1} \;\approx\; \sum_a \widehat{q}_a^{\,2}.$$

The approximation improves with N. Real language also has dependencies between positions.

APPENDIX

A10 / Score a candidate using all letter counts

Let $O_a(k)$ count plaintext letter a after subtracting candidate shift k from the column.

$$E_a = N q_a, \qquad \chi^2(k) = \sum_{a=0}^{25} \frac{(O_a(k) - E_a)^2}{E_a}.$$

The q_a values form an illustrative English frequency model, normalized to sum to 1.

A smaller score means a closer fit to that model. This is a ranking heuristic, not proof that a key is correct.

The full model probabilities are included in Appendix A4.

APPENDIX

A11 / Work out several score contributions

Plaintext a	Observed $O_a(11)$	Expected Nq_a	Contribution
E	27	26.547	0.008
N	20	14.106	2.463
A	18	17.069	0.051
Z	0	0.155	0.155

$$\chi^2(11) = \sum_{a=0}^{25} \text{contribution}_a = 22.297.$$

Sum all 26 terms. The four rows above are selected contributions, not the complete sum.

APPENDIX

A12 / Compare the candidate shifts

Shift k	Key letter	Full χ^2 score
11	L	22.297
24	Y	526.998
22	W	699.130
17	R	839.822

Shift 11 gives the smallest score among all 26 candidates for this column.

Repeat the same calculation independently for the other four columns.

This avoids enumerating the entire product of 26^5 possible five-letter keys.

APPENDIX

A13 / Recover all five key letters

Key index β (0–4)	Best shift	Key letter	Full χ^2 score
0	11	L	22.297
1	4	E	29.388
2	12	M	28.459
3	14	O	23.656
4	13	N	23.953

The resulting key hypothesis is LEMON. We must now test it on the interleaved complete message.